

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer who has led 20+ zero-downtime data migrations off legacy systems — and built the safe-import machinery that made them repeatable: real-time two-way sync with rollback held open for the entire cutover window, idempotent message processing, validation of messy production data, and audit-friendly phased cutovers. At MTailor (US, remote) I owned the company-wide Parse/MongoDB → Firebase/GCP migration end to end, reporting directly to the CTO, and cut \$5,000/month in infrastructure costs. Now applying the same discipline to migration platforms: turning one-off legacy-system cutovers into automated, auditable pipelines — with pragmatic use of LLM APIs for extraction, classification, and matching.

CORE COMPETENCIES

Data Migration & ETL

Zero-Downtime Cutovers

Python

Node.js / TypeScript

SQL & NoSQL Data Modeling

Validation & Deduplication

External API Integrations

LLM APIs & AI Tooling

Docker / Linux / GCP

Audit-Safe Import Flows

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Led the end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO — owning architecture, implementation, testing, deployment, and production support.
- Built a **real-time two-way sync between MongoDB and Firestore** (Node.js + Google Pub/Sub) with idempotent, redelivery-safe processing — keeping both datastores consistent under live traffic and preserving rollback for every service through the entire cutover window.
- Wrote migration scripts that **extracted, transformed, and validated messy production data** from a poorly documented legacy platform, running complex processing logic before safe import into the new system.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a **Python** script; moved all remaining services off AWS, **saving \$5,000/month**.
- Shipped revenue features end to end: Express Shipping (+\$40 revenue per order) and a 3D visualisation feature (ffmpeg video overlay) that increased buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Migration Contractor — US Remote

- Hired via Upwork for a complex legacy-data migration; delivered MongoDB→Firestore migration tooling on an unfamiliar stack, then **wrote the Firebase SDK documentation (Node.js and Python)** that other engineers onboarded from.
- Trained the Ops team on the new dashboard and backup procedures — converted to full-time and given ownership of the company-wide migration within months.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Built batched ad-hoc jobs querying **2M+ Firestore records** to keep customer stock prices current from third-party Morningstar APIs — optimizing query shape over hardware for cost and latency.
- Extended backend REST APIs with Firebase Cloud Functions and Express.js; shipped responsive PWA pages in React and Tailwind.

CodeBits

Jan 2020 – Jul 2021

Software Engineer (Team Lead) — Uganda Remote

- Led a team of 4 building legal-tech systems for NGOs: case management for FIDA Uganda and a paralegal database for LASPNET.
- Architected an **event-driven microservices backend** (Apache Kafka, Docker, Kubernetes) and a USSD service integrating multiple external providers against one versioned gRPC contract.

SELECTED PROJECTS

Marketplace Recommendation & Matching Engine

Collaborative + content-based filtering engine matching users to service providers across a messy, duplicate-prone catalog — ranked confidence scoring instead of binary matching.

MongoDB, Node.js

Healthcare WhatsApp Chatbot with Prescription OCR

Production LLM integration: OpenAI + Google Vision extracting structured data from prescription images via the WhatsApp Business API — safety-sensitive extraction with human review.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Career-Ops — AI-Assisted Evaluation Pipeline (open source)

Agentic pipeline for evaluating and classifying job postings at scale: live source verification, structured extraction, automated document generation — AI coding tools and LLM APIs in daily production use.

Node.js, Claude Code, Playwright, YAML

EDUCATION

BSc Computer Engineering — [Makerere University](#)

2018 – 2022

SKILLS

Proficient: Python, SQL, Node.js, MongoDB, Firebase/Firestore, GCP (Pub/Sub, GCS, Cloud Functions), Linux, Git, React, HTML/CSS

Intermediate: Go, AWS, Rust, React Native **Tooling:** Docker, Kubernetes, Kafka, gRPC, LLM APIs (OpenAI, Claude), AI coding workflows

Languages: English, Kiswahili